

Computing Admissible Strings II

Finite automata, nondeterminism, conversions, and transducers

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Finite-state computation remembers only a finite abstraction of the past

Question

Which distinctions about an input prefix must a machine retain to decide the whole string?

$$M = (Q, \Sigma, \Delta, s, F)$$

MOLR §§5.2, 5.4

Where we are

1. Finite automata
2. Subset construction
3. Conversions
4. Transducers

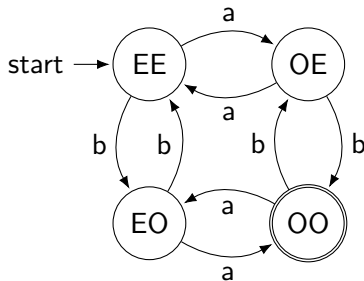
A state is an equivalence class of relevant histories

- Two prefixes may lead to the same state if no possible continuation needs to distinguish them.
- A transition updates that finite summary after one input symbol.
- Acceptance depends on the final summary and exhausted input.

Take-away

States do not store the prefix; they preserve only what future decisions require.

Example: odd numbers of both a 's and b 's



Guided analysis

Trace *abba*. What invariant does each state name encode?

MOLR §5.2

Determinism is a property of the transition relation

DFA

For each state and input symbol, exactly one next state is available.

NFA

Zero, one, or several next states may be available; ϵ -moves may occur.

Take-away

Nondeterminism changes the description of the computation, not the class of regular languages recognized.

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A DFA state can represent a set of NFA possibilities

- 1 Start with the ϵ -closure of the NFA start state.
- 2 For each input symbol, collect every reachable NFA state.
- 3 Take the ϵ -closure again.
- 4 Make each distinct set a DFA state.
- 5 Mark a set accepting when it contains an NFA accepting state.

MOLR §5.4, eliminating nondeterminism

Subset construction preserves acceptance, not the original path story

- An NFA may accept because at least one path succeeds.
- The DFA tracks all currently possible NFA states simultaneously.
- The resulting state names are sets, often including unreachable candidates that can be removed.

Guided analysis

If an NFA has states $\{s, p, q\}$, what are the possible subset-state names? Must all occur?

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Regular grammars and finite automata encode the same stringsets

For a right-linear rule

$$A \rightarrow aB$$

create a transition

$$A \xrightarrow{a} B.$$

For $A \rightarrow a$, transition on a to a designated accepting state.

- Nonterminals become states.
- Derivational continuation becomes machine continuation.

Automaton-to-regexp conversion eliminates internal states

- Relax edges to carry regular expressions rather than single symbols.
- Combine parallel alternatives with union.
- Compose incoming and outgoing paths when eliminating a state.
- Preserve loops with Kleene star.

Take-away

Conversion preserves the admitted strings while changing which regularities are visually salient.

MOLR §5.4, regular expressions and automata

Where we are

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A finite-state transducer computes a regular relation

Edges pair input with output:

$$q \xrightarrow{a:b} p, qquad q \xrightarrow{a:\epsilon} p, qquad q \xrightarrow{\epsilon:b} p.$$

- $a : b$: correspondence; $a : \epsilon$: deletion; $\epsilon : b$: insertion.
- A path relates the concatenated input and output labels.
- Functionality is an extra condition, not guaranteed by transduction.

MOLR §5.4, transducers

Composition makes finite-state analyses modular

$$R_1 \subseteq \Sigma^* \times \Gamma^*, \quad R_2 \subseteq \Gamma^* \times \Delta^*$$

$$R_2 \circ R_1 \subseteq \Sigma^* \times \Delta^*.$$

- The intermediate alphabet must connect.
- Composition hides intermediate representations while preserving their contribution.
- This is the same domain condition encountered in Week 1.

Third hour: follow, determinize, convert

Given a small NFA supplied in class:

- ① Trace two accepted and two rejected strings.
- ② Calculate each needed ϵ -closure.
- ③ Construct reachable DFA subset states.
- ④ Translate the result to a right-linear grammar.
- ⑤ Add output labels to express one deletion relation.

Week 10 synthesis

- ① States preserve finite distinctions among input histories.
- ② NFAs and DFAs differ as computations but recognize the same family.
- ③ Subset construction makes concurrent possibilities explicit.
- ④ Regular grammars, regexps, and automata are interconvertible descriptions.
- ⑤ Transducers compute regular relations between strings.

Question

Quiz 8 asks for one trace, closure, subset step, or conversion.